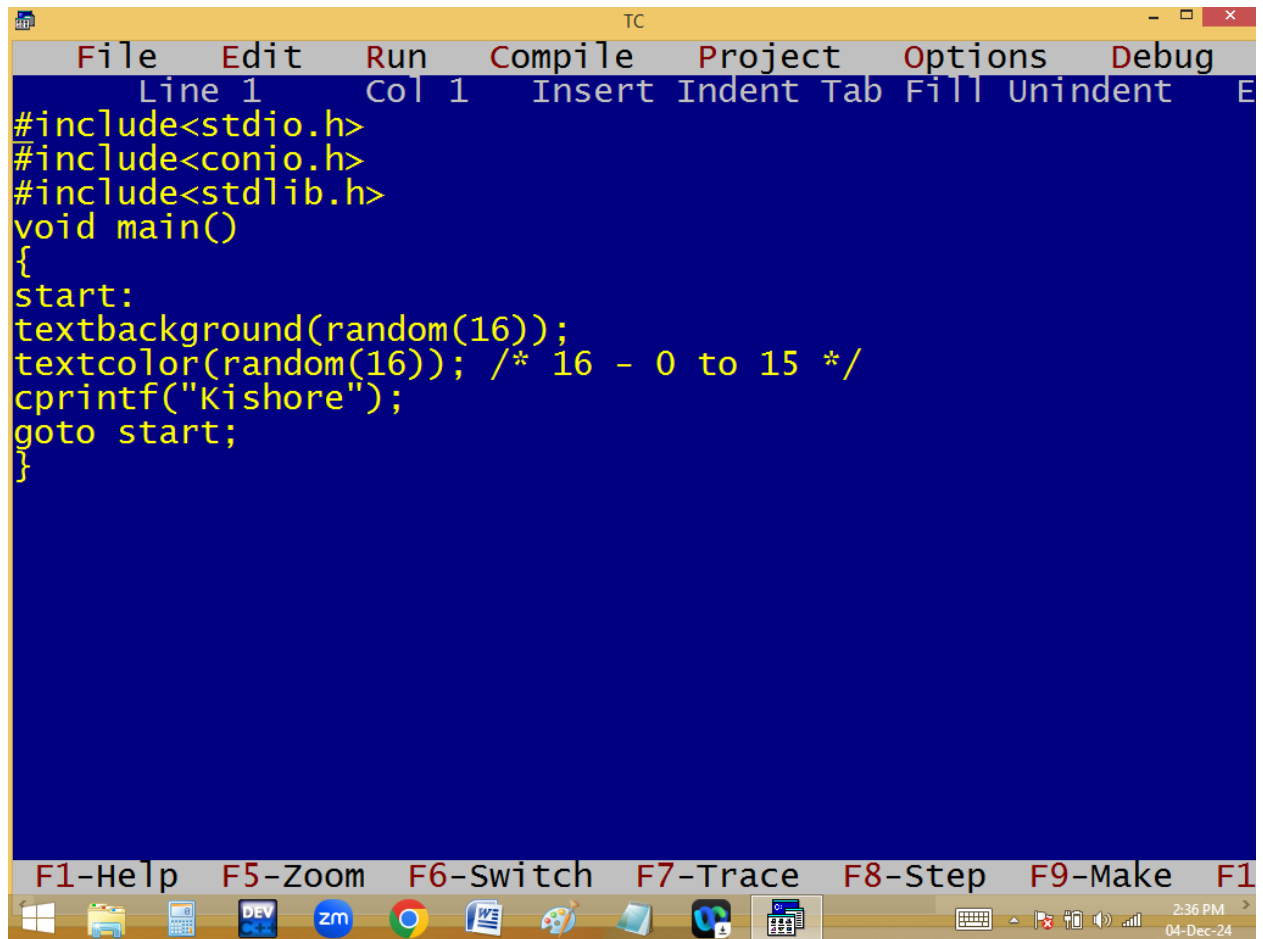
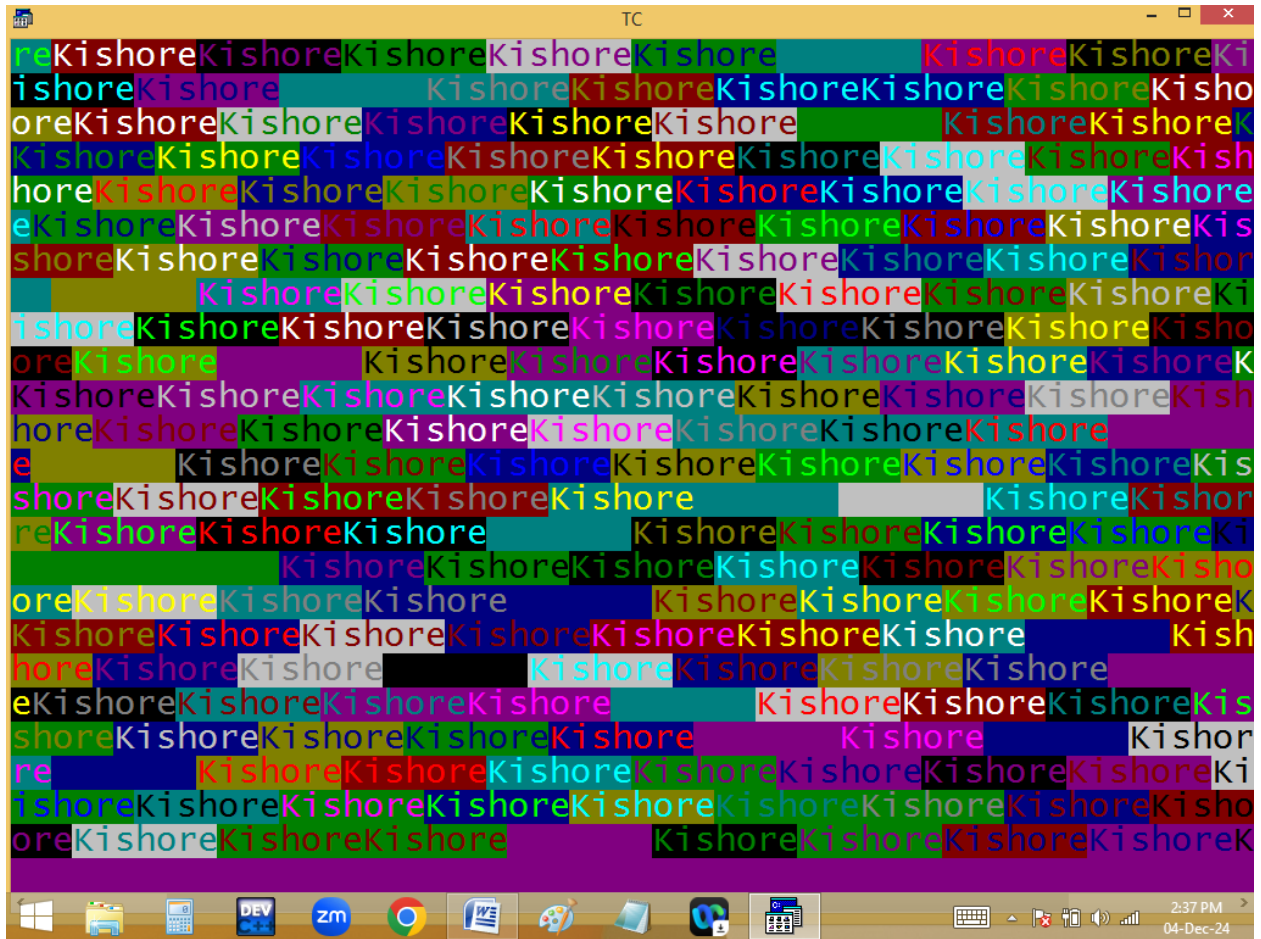




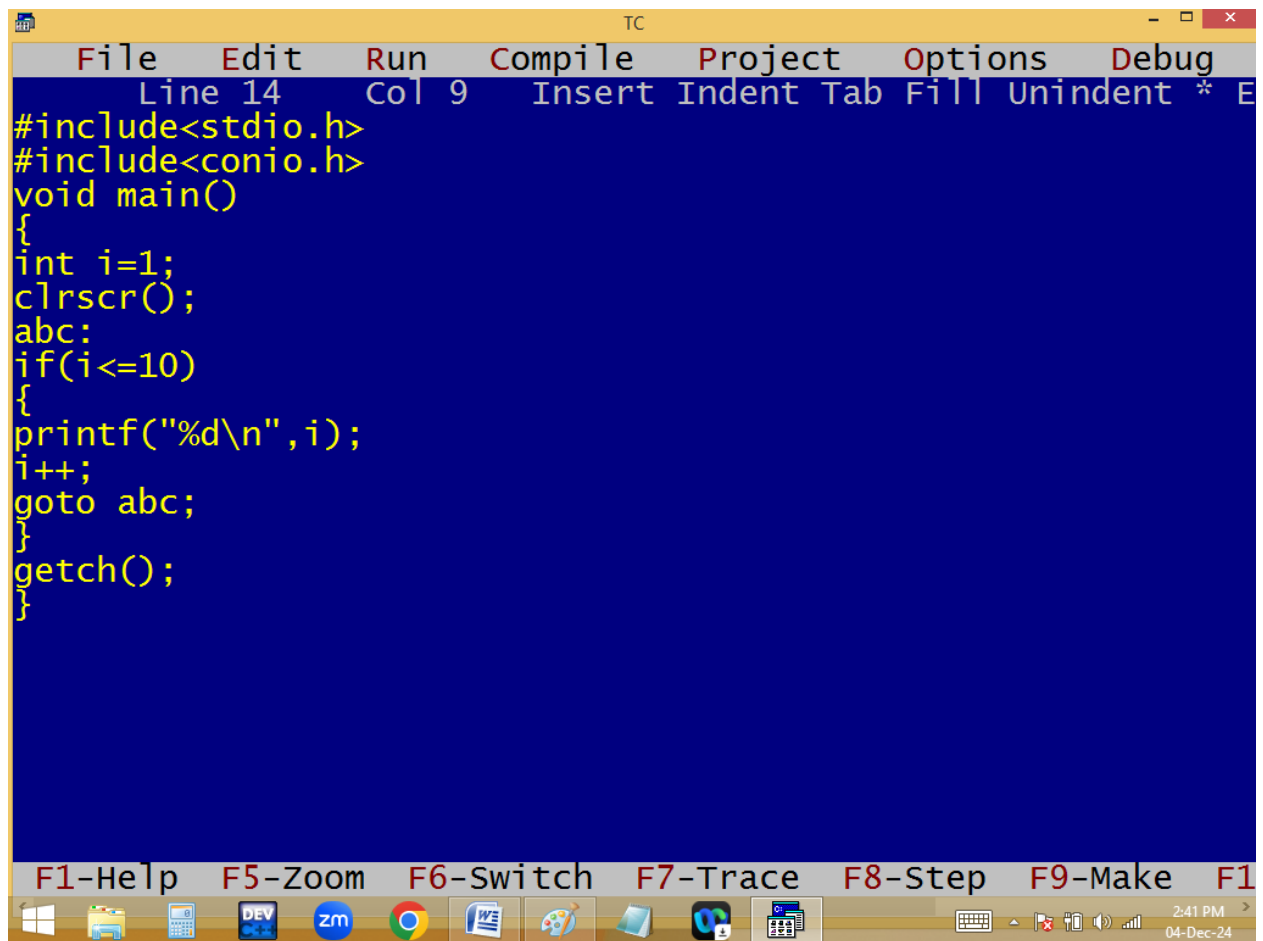
The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top shows "Line 1", "Col 1", and "Insert Indent Tab Fill Unindent". The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
start:
textbackground(random(16));
textcolor(random(16)); /* 16 - 0 to 15 */
cprintf("Kishore");
goto start;
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the bottom, showing icons for File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner shows "2:36 PM" and "04-Dec-24".



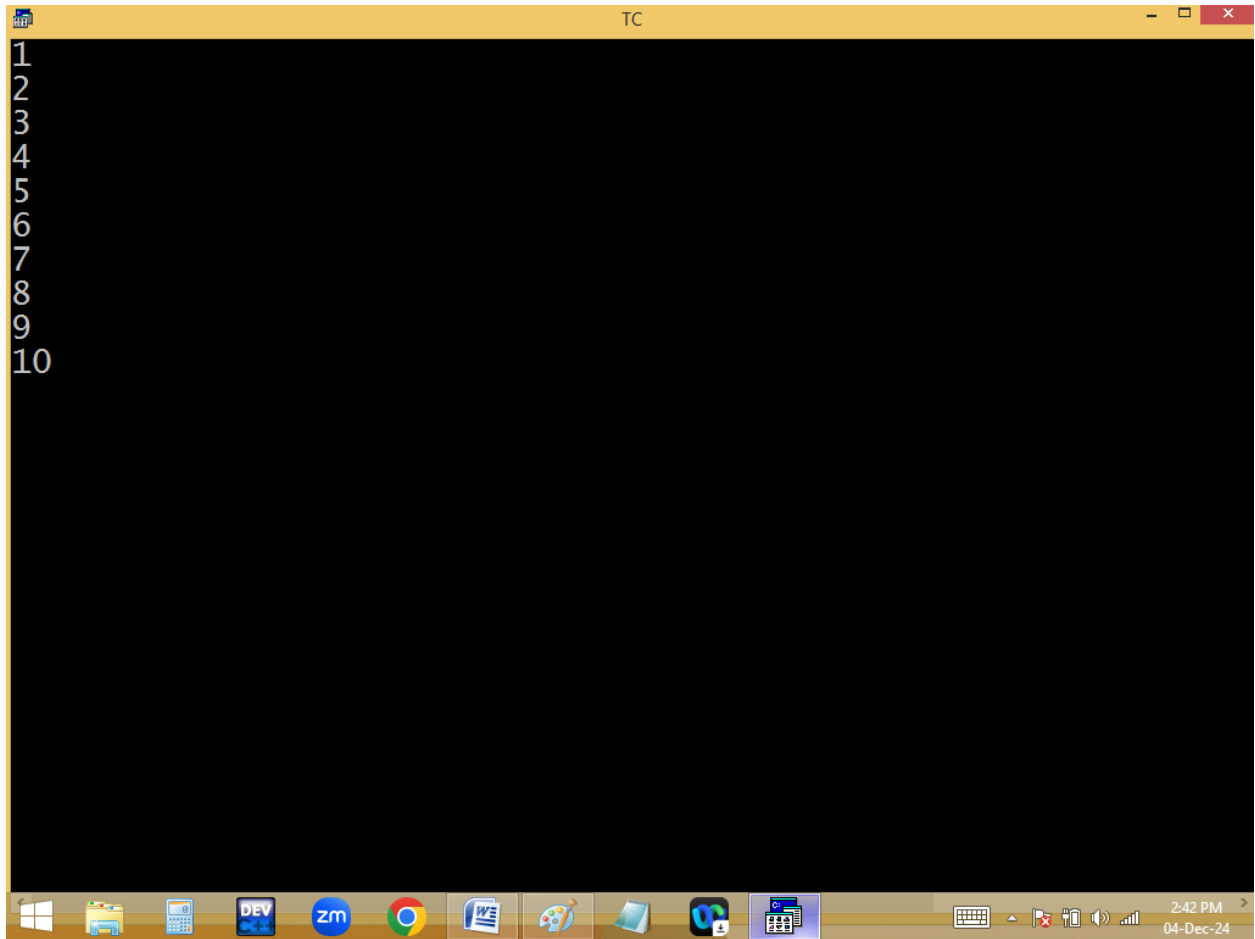
Write a c program to print 1..10 no's without using loop:

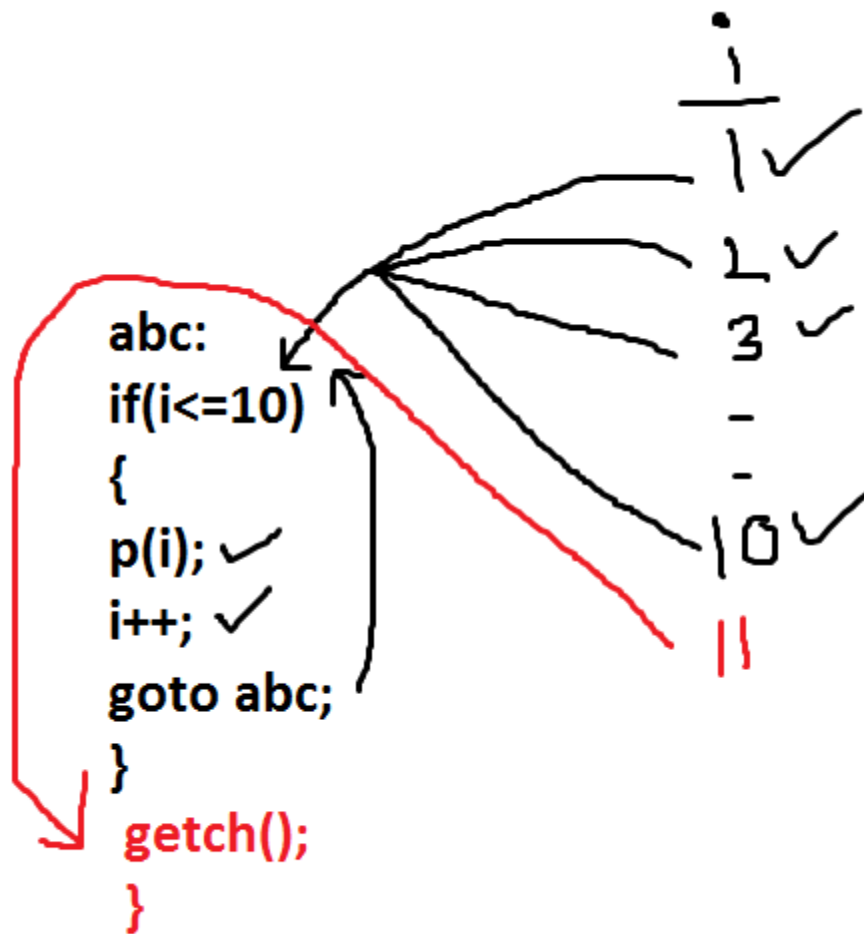


The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar displays "Line 14", "Col 9", and various editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a "Breakpoint" icon. The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
abc:
if(i<=10)
{
printf("%d\n",i);
i++;
goto abc;
}
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions. Above the toolbar, a row of function key shortcuts is visible: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows several application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder icon. The system tray on the right side of the taskbar shows the time as "2:41 PM" and the date as "04-Dec-24".





switch:

It is a selection statement.

It is used to execute one case of statements from no of cases according to the switch expression value matched with case expression value. In switch the program is jumped to matching case like the go to label.

It is similar to ladder if in working style.

Switch performance is high when compared with ladder if because of it jumps to matching case.

Syntax:

```
switch(condition / expression)  
{  
  case constexp1:  
    statements;  
    break;
```

```
case  constexp2:
statements;
break;
case  constexpN:
statements;
break;
[ default: statements; ]
}
```

Here switch, case, break, default are the keywords.

In between case and case expression / value at least one space should be provided. Otherwise it will become a label.

case expression/value should be a constant integer/char value. i.e. float / string not allowed.

One case contains one expression only.

case expression doesn't contain any separators like , . etc.

case expression should be end with : (colon)

Each case should be separated with break keyword. Otherwise remaining cases also executed.

Duplicate cases not allowed.

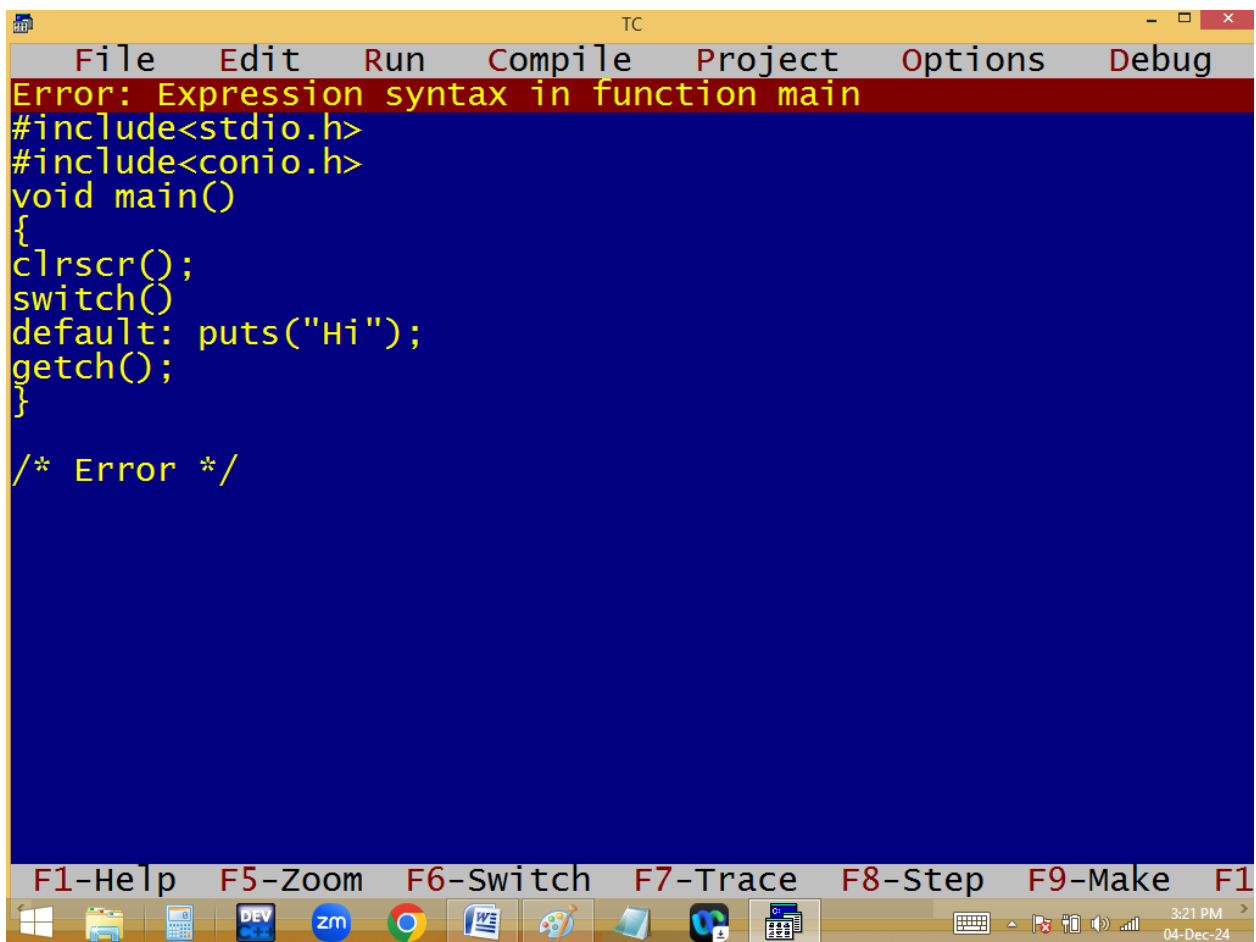
default is similar to the else and all cases are failed then default statements are executed. Default is optional and we can declare it anywhere in our switch.

Outside case expressions not considered in switch.

Note:

- 1. Case value should be a const int.**
- 2. Variables not allowed in case.**
- 3. Float / string not allowed in case.**
- 4. Duplicate cases not allowed.**
- 5. Case and case value have to be separated with at least one space.**
- 6. If space is not given, it is converted into a label.**
- 7. Every case should be end with :**
- 8. break is optional and when it is not there, remaining case also executed until the next break is occurred.**

9. Case doesn't allow range of values.
10. Case doesn't allow , . like separators,
11. Default is optional and working similar to else. We can declare default anywhere.
12. Switch, case, break and default are the keywords.
13. Outside case expressions not considered.

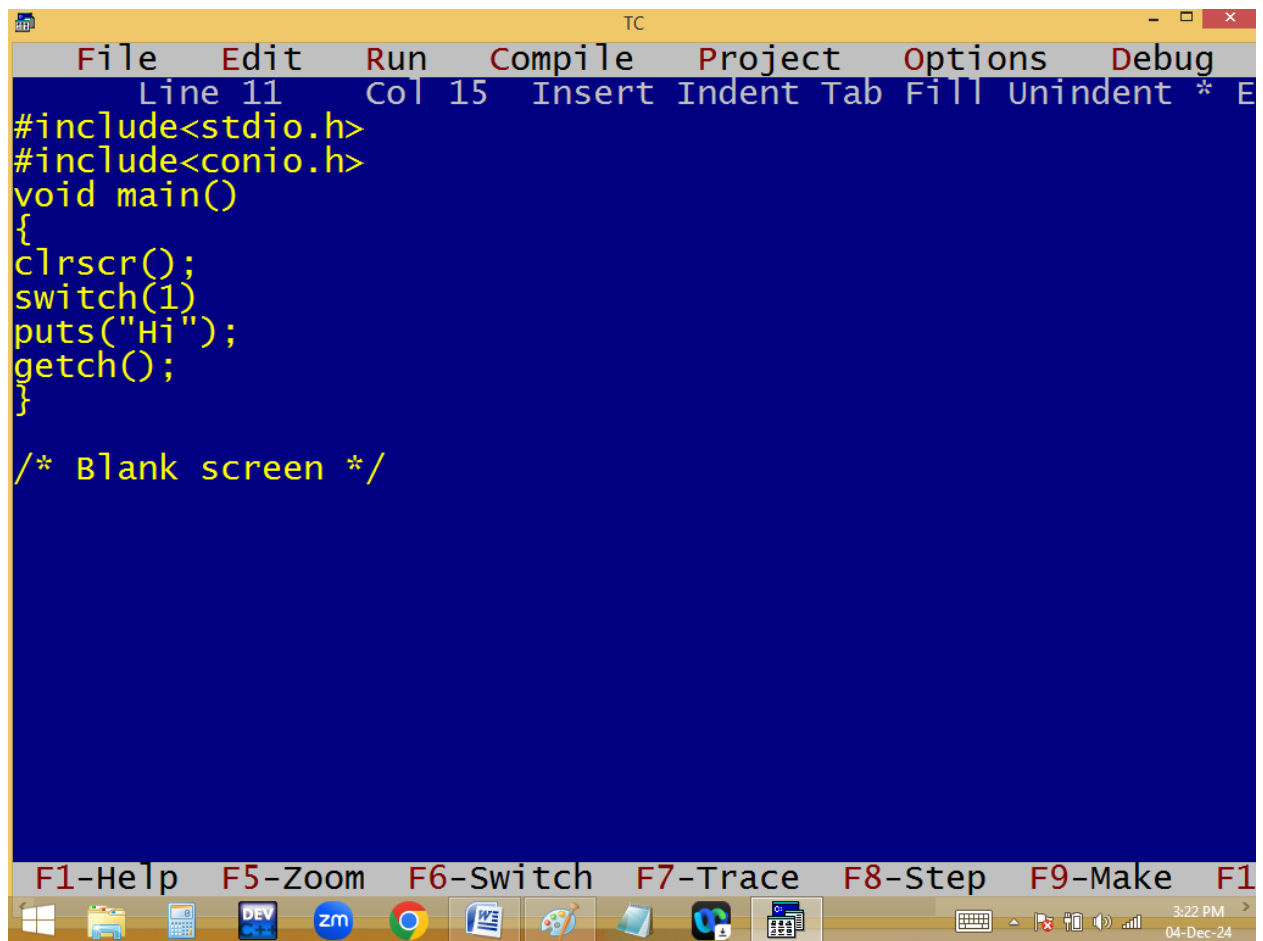


The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. A red error banner at the top reads "Error: Expression syntax in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch()
default: puts("Hi");
getch();
}

/* Error */
```

The bottom status bar displays function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the bottom shows the system clock as 3:21 PM on 04-Dec-24.

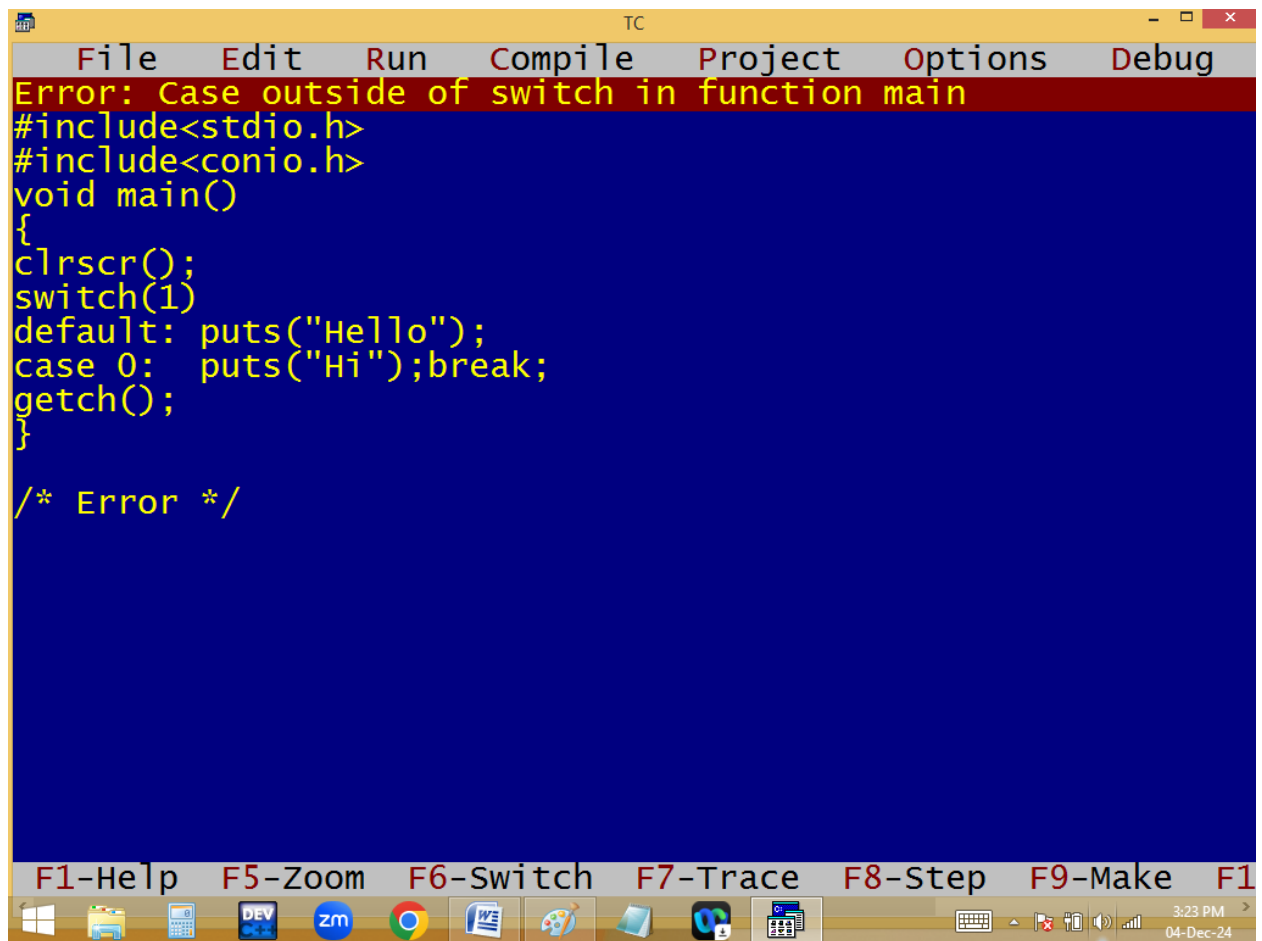


The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 11", "Col 15", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
puts("Hi");
getch();
}

/* Blank screen */
```

Below the code editor, there is a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner shows "3:22 PM" and "04-Dec-24".

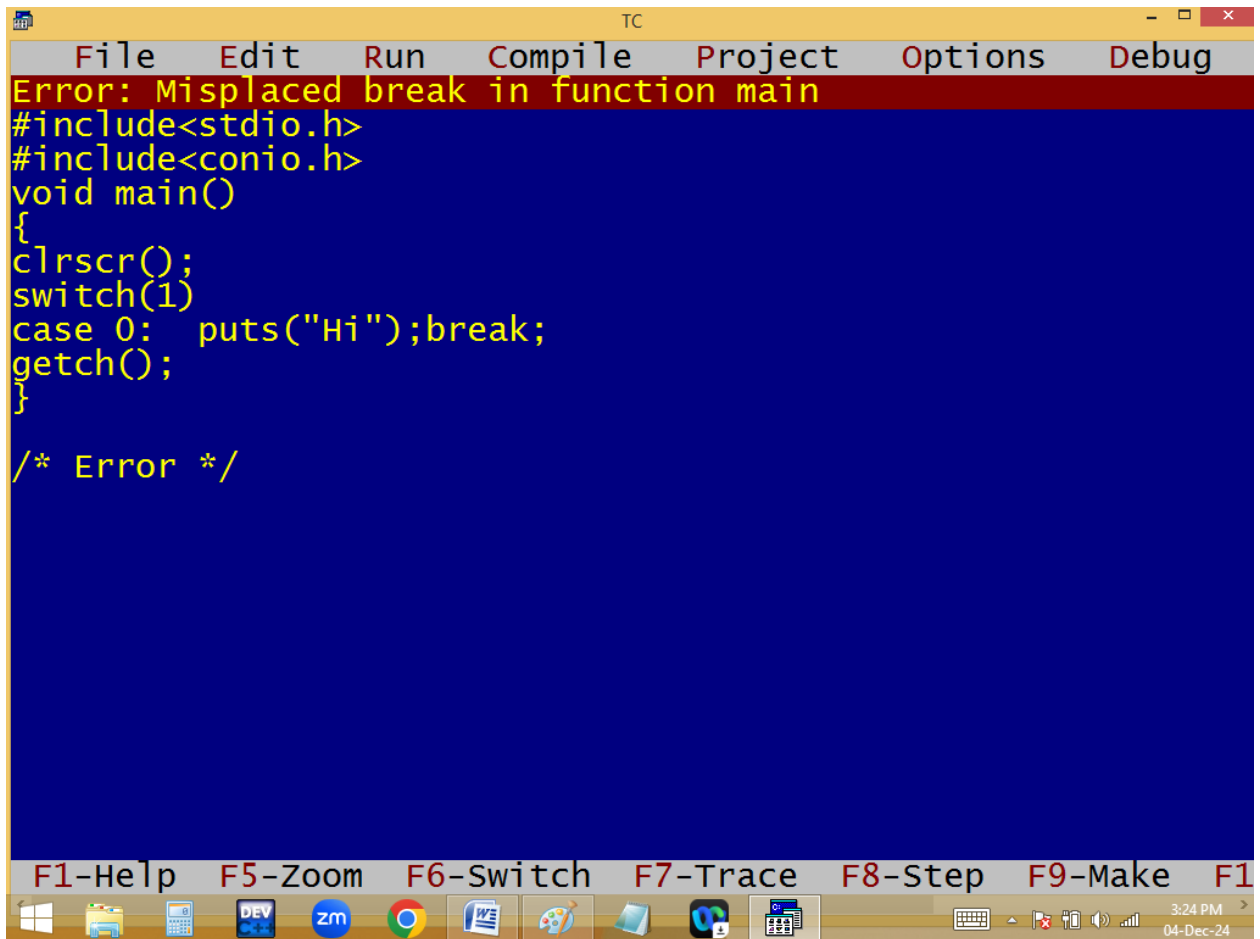


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Case outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
default: puts("Hello");
case 0:  puts("Hi");break;
getch();
}

/* Error */
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner displays "3:23 PM" and "04-Dec-24".

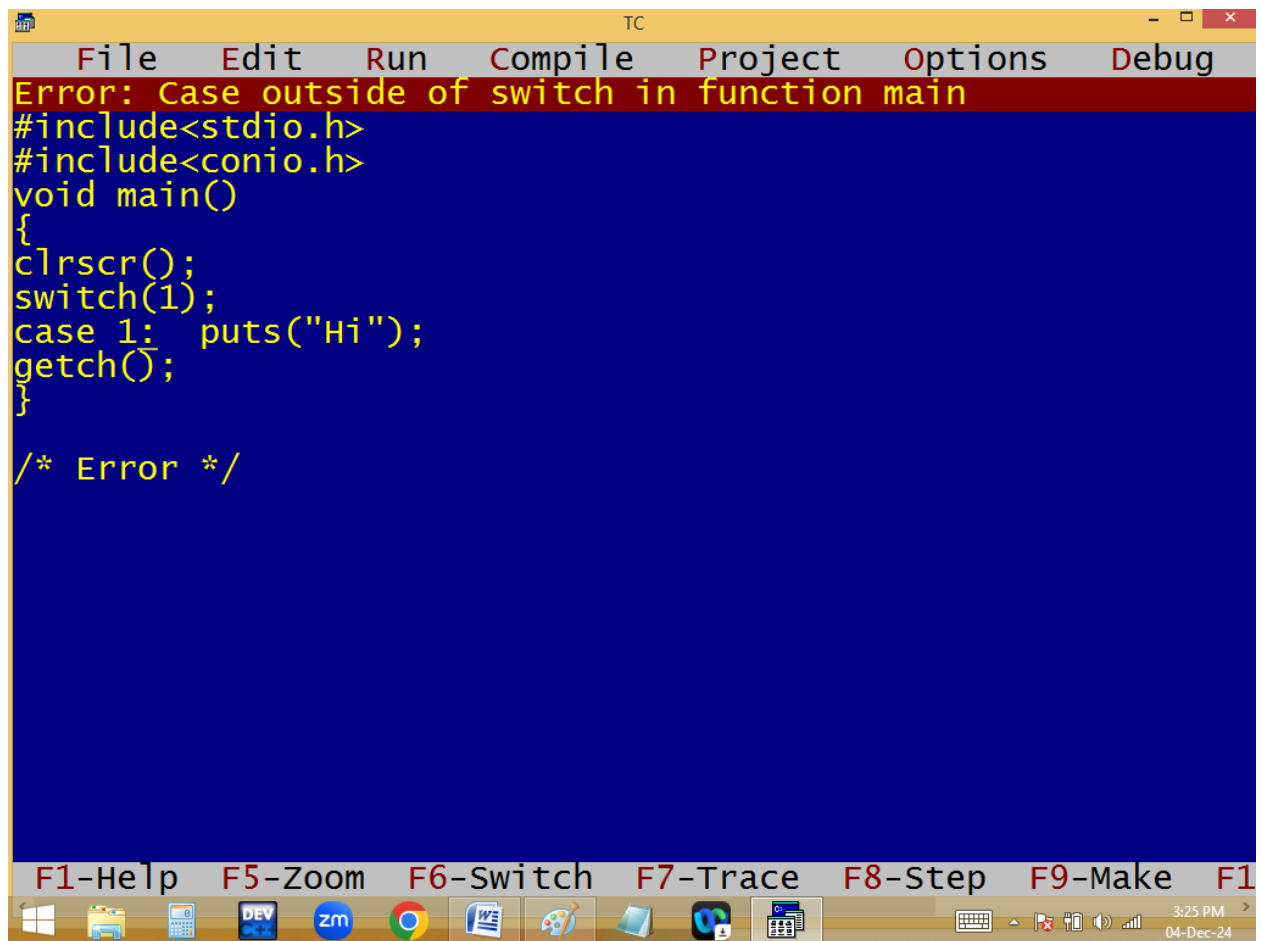


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code editor displays the message "Error: Misplaced break in function main". The code editor has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
case 0:  puts("Hi");break;
getch();
}

/* Error */
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "3:24 PM" and "04-Dec-24".

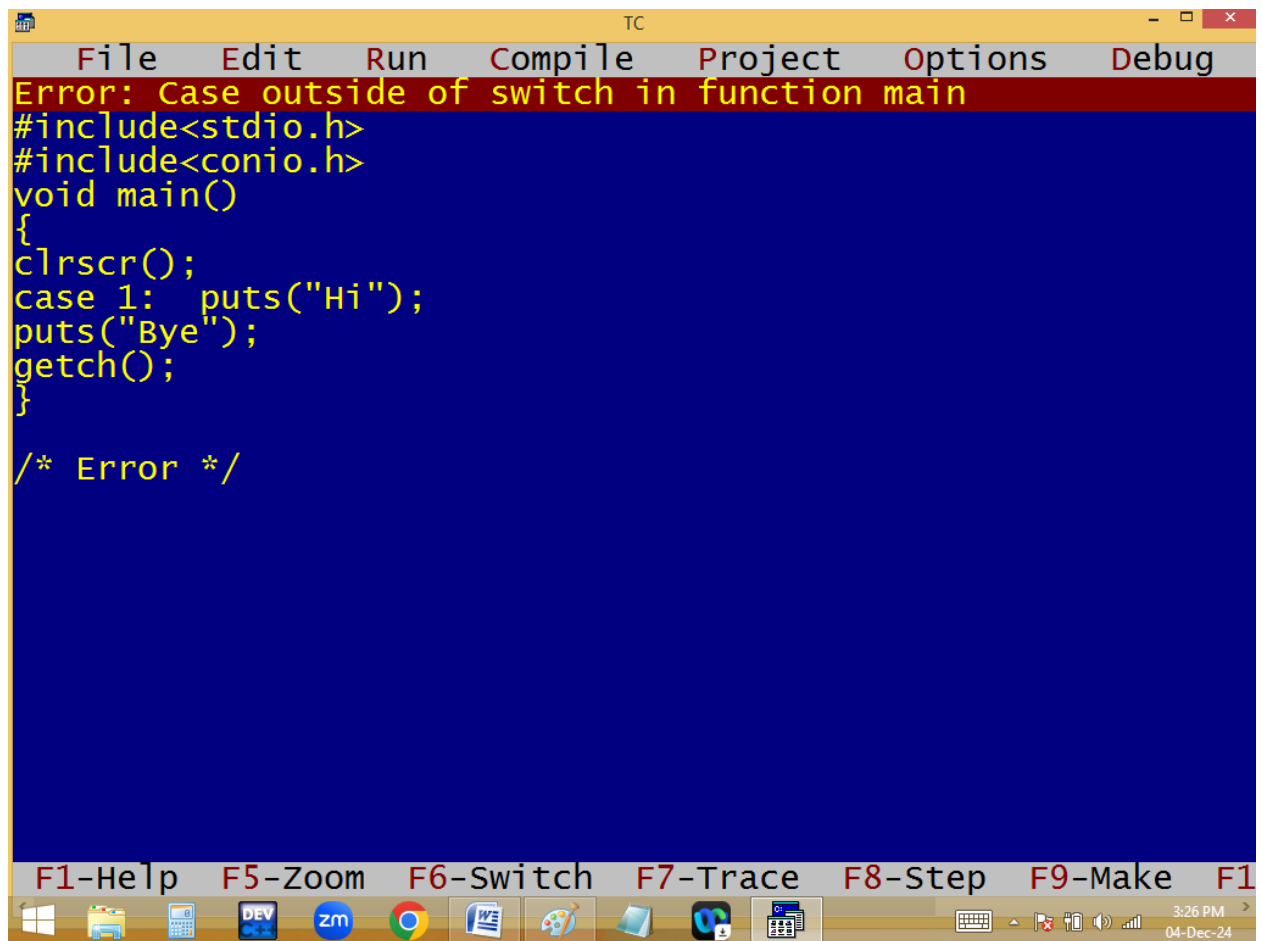


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code area states: "Error: Case outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1);
case 1: puts("Hi");
getch();
}

/* Error */
```

Below the code area, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner displays "3:25 PM" and "04-Dec-24".

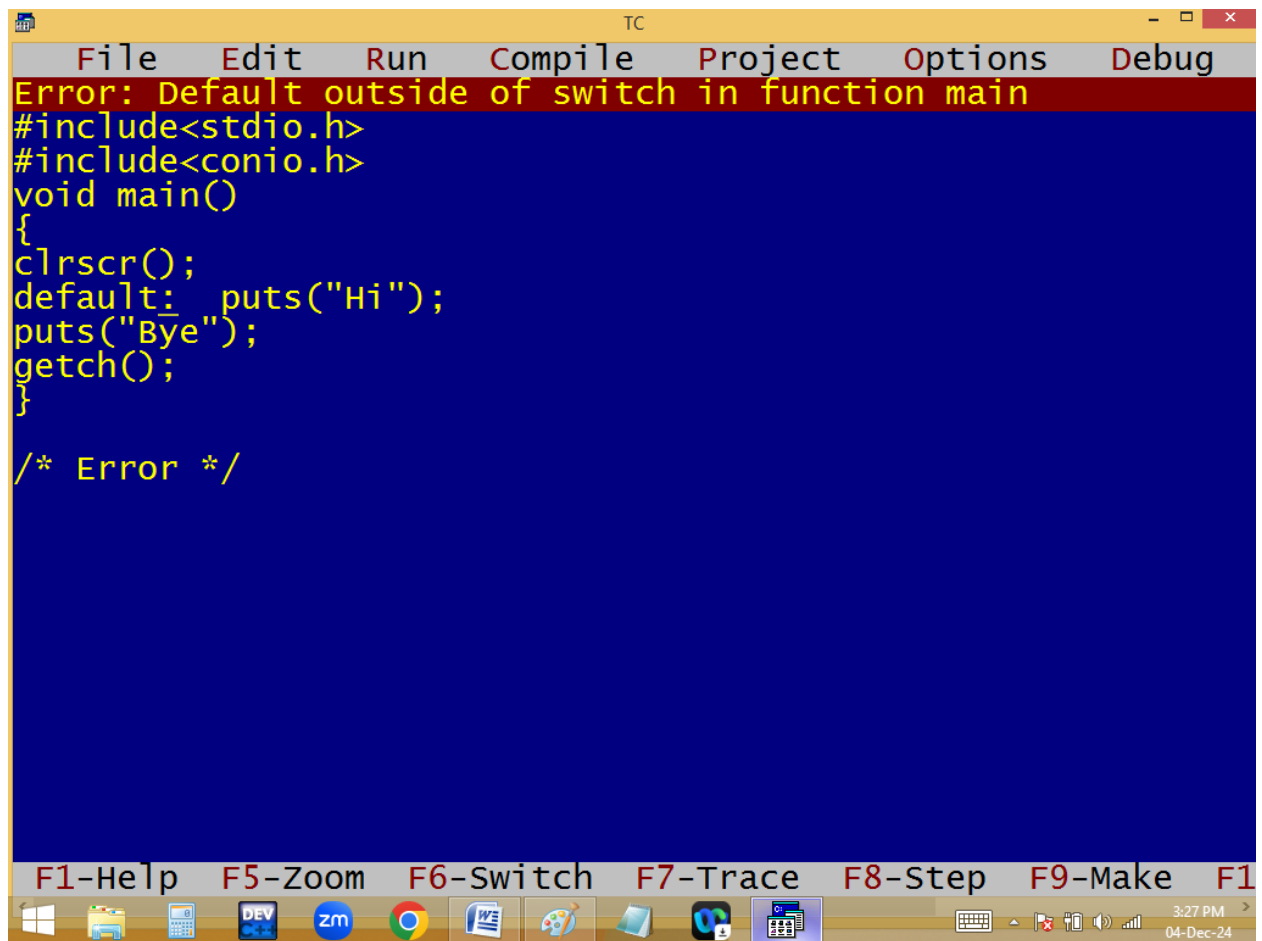


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Case outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
case 1: puts("Hi");
puts("Bye");
getch();
}

/* Error */
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system clock in the bottom right corner displays "3:26 PM" and "04-Dec-24".

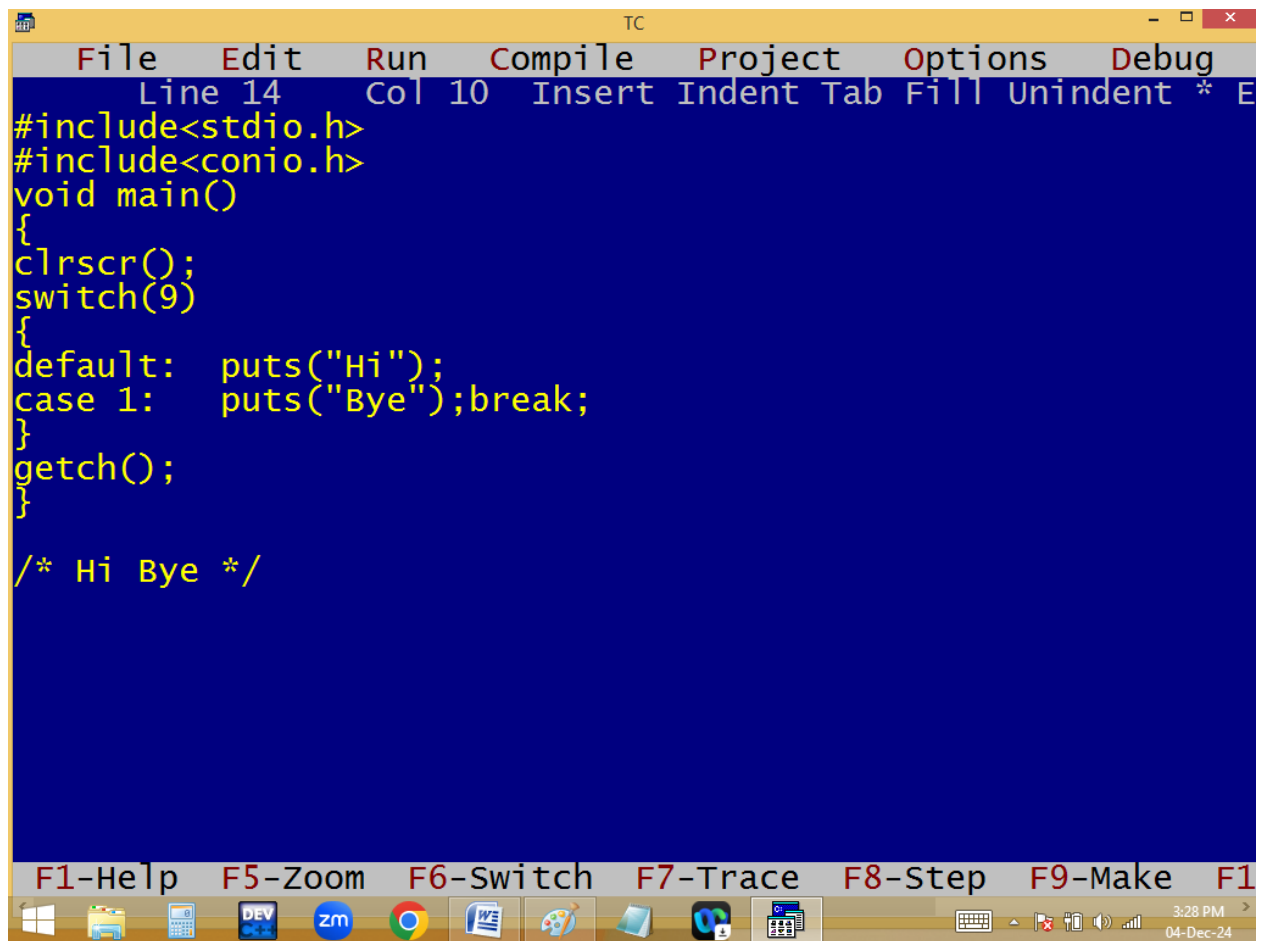


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the editor area states: "Error: Default outside of switch in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
default: puts("Hi");
puts("Bye");
getch();
}

/* Error */
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and others. The system clock in the bottom right corner indicates the time is 3:27 PM on 04-Dec-24.

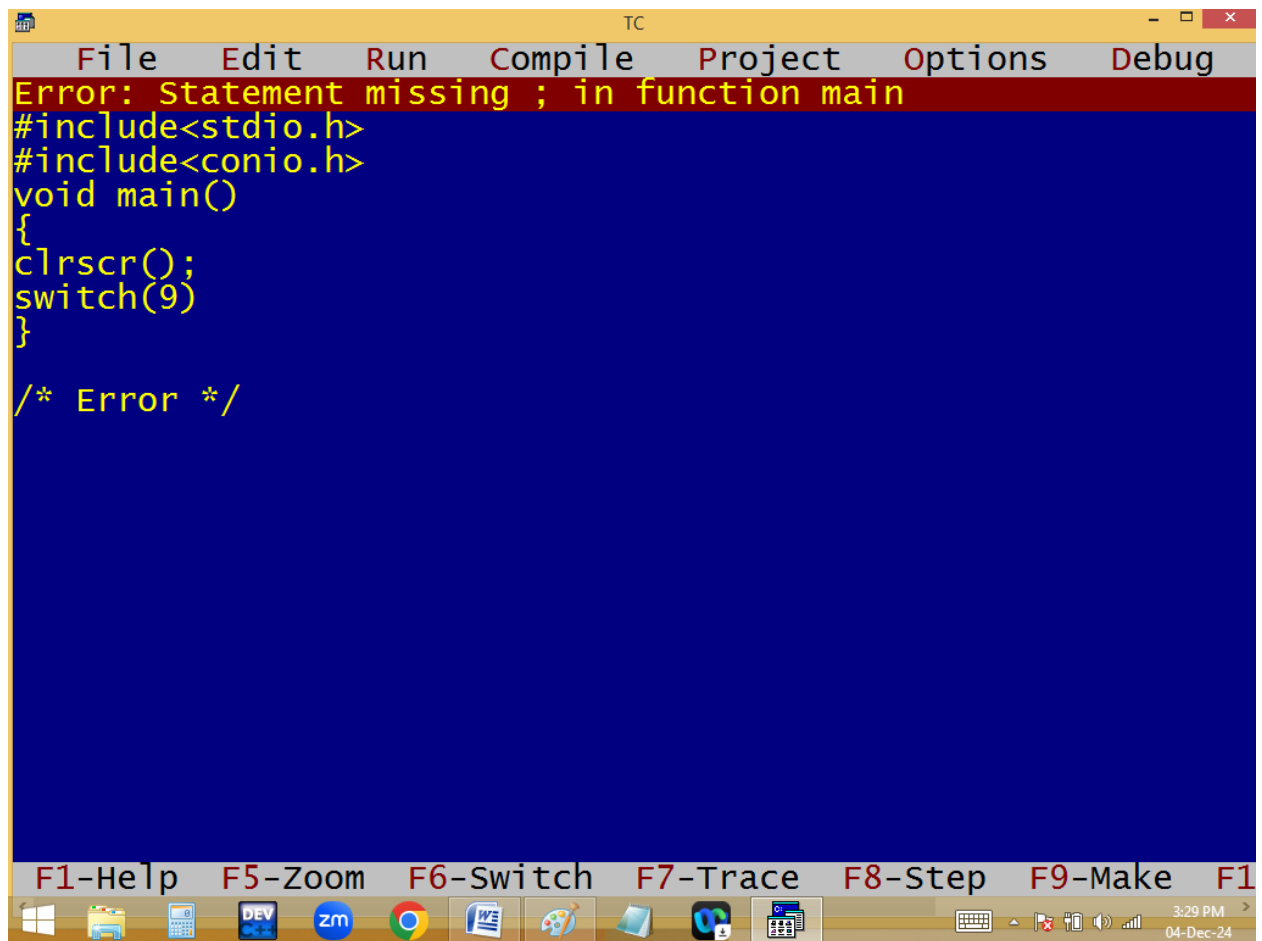


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 14 Col 10" and provides keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(9)
{
default:  puts("Hi");
case 1:   puts("Bye");break;
}
getch();
}

/* Hi Bye */
```

At the bottom of the window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and other applications. The system clock in the bottom right corner shows "3:28 PM" and "04-Dec-24".

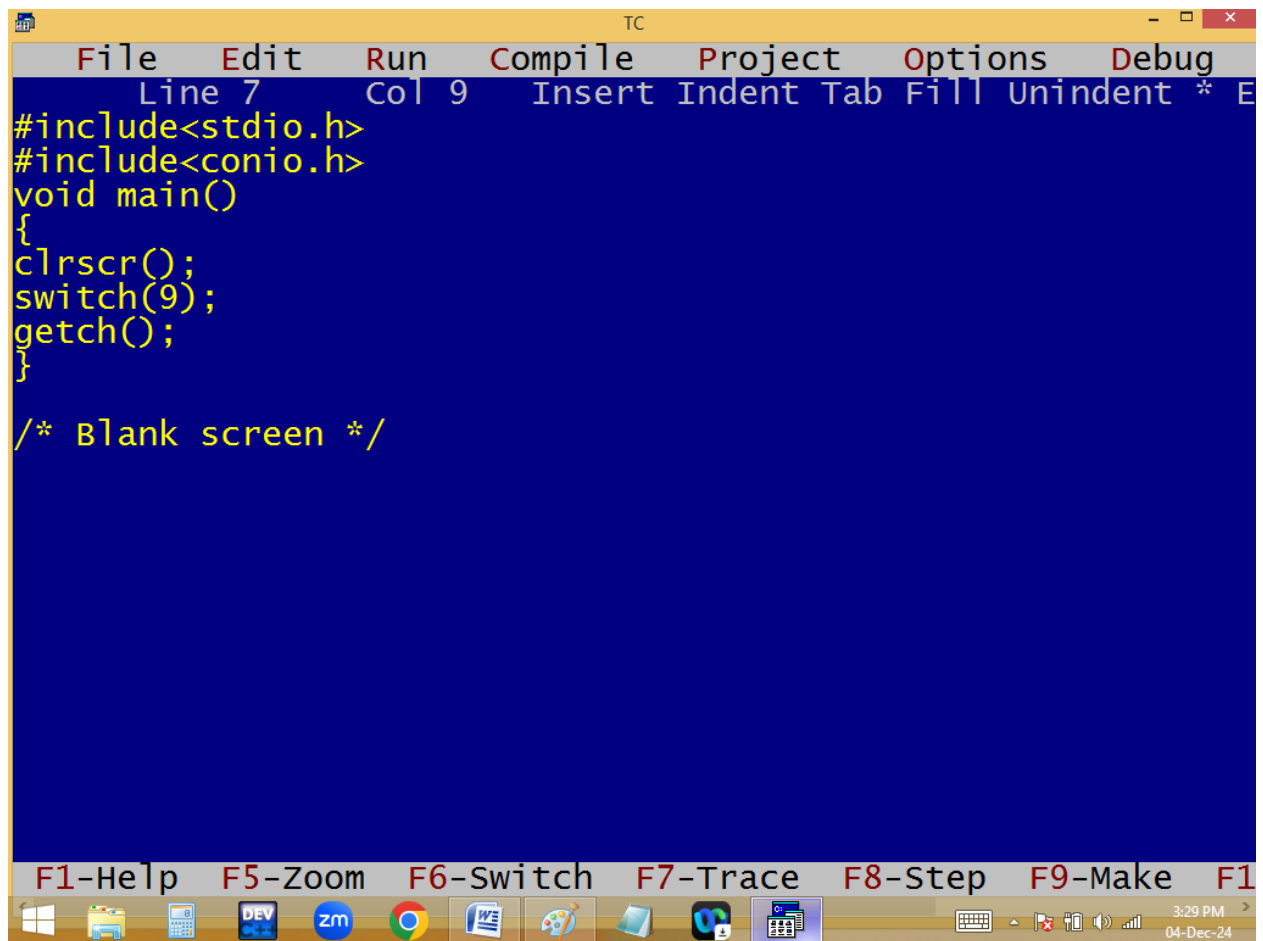


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the editor area states "Error: Statement missing ; in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(9)
}

/* Error */
```

Below the editor, a toolbar contains function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom shows various application icons, including File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock in the bottom right corner displays "3:29 PM" and "04-Dec-24".

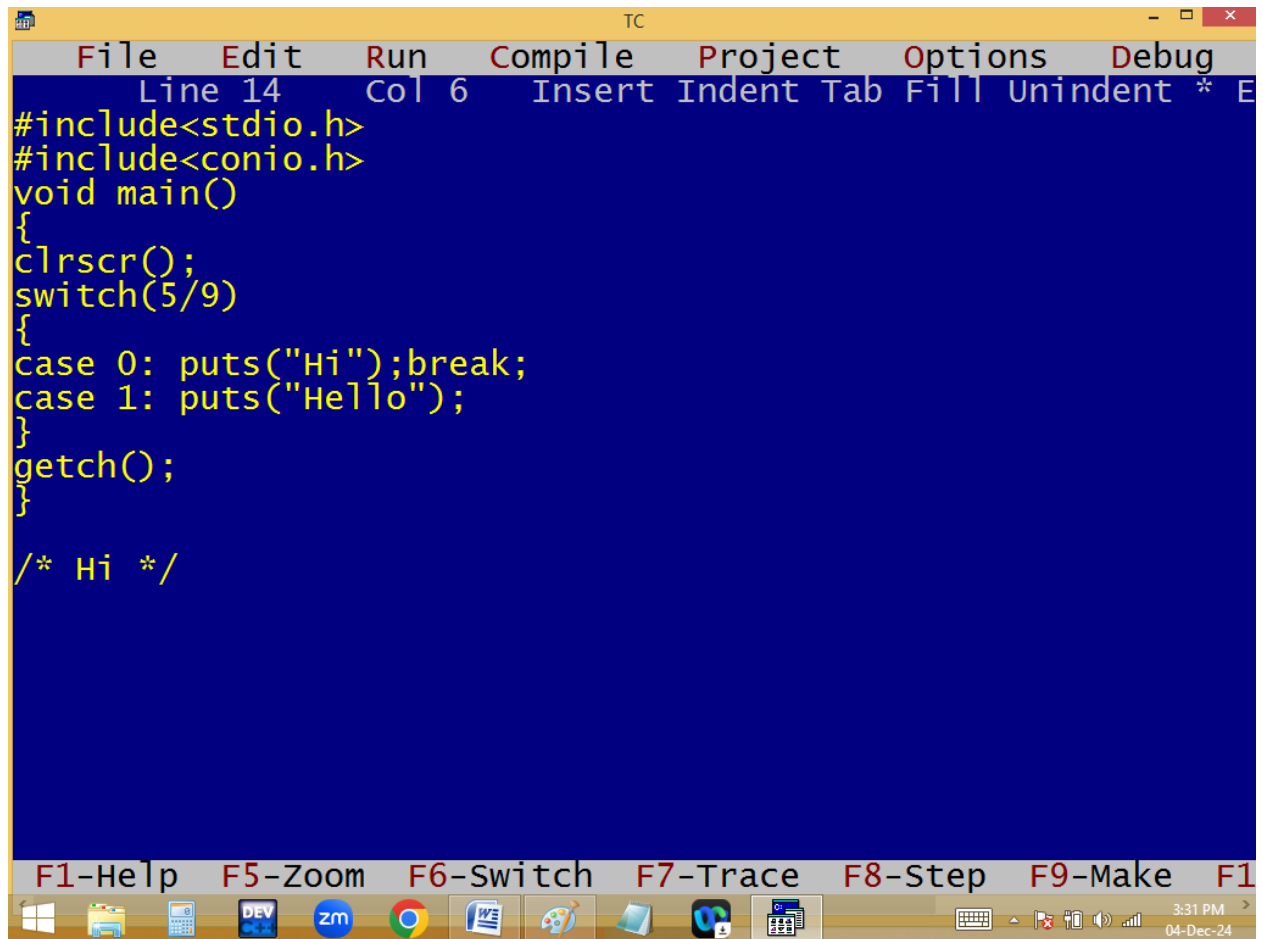


The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 7", "Col 9", and lists editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(9);
getch();
}

/* Blank screen */
```

Below the code editor, there is a row of function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time as 3:29 PM and the date as 04-Dec-24.

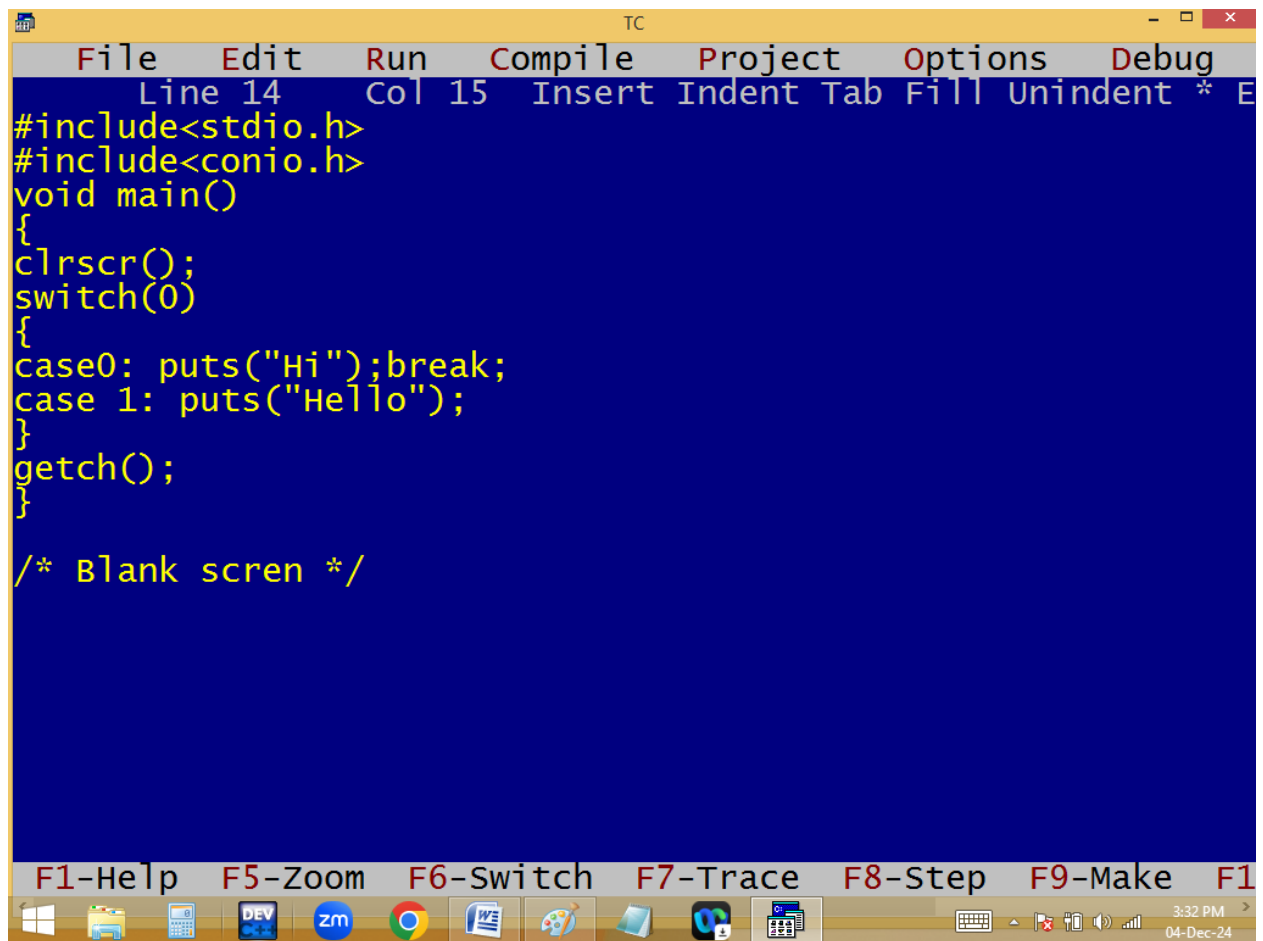


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 6", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(5/9)
{
case 0: puts("Hi");break;
case 1: puts("Hello");
}
getch();
}

/* Hi */
```

The bottom of the window features a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and others. The system clock in the bottom right corner displays "3:31 PM" and "04-Dec-24".

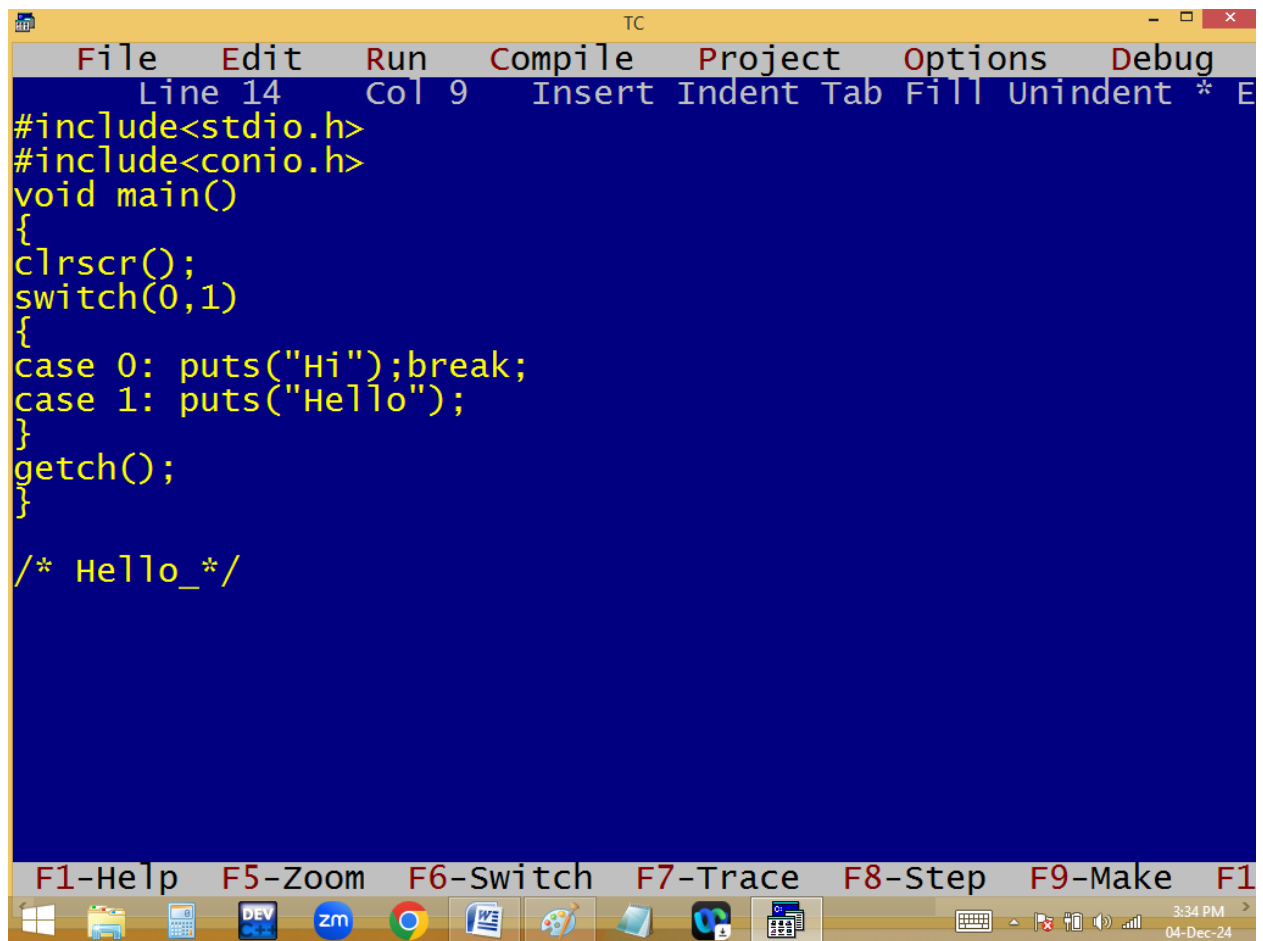


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 15", and "Insert" mode. The code editor has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(0)
{
case0: puts("Hi");break;
case 1: puts("Hello");
}
getch();
}

/* Blank screen */
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system clock shows "3:32 PM" and "04-Dec-24".

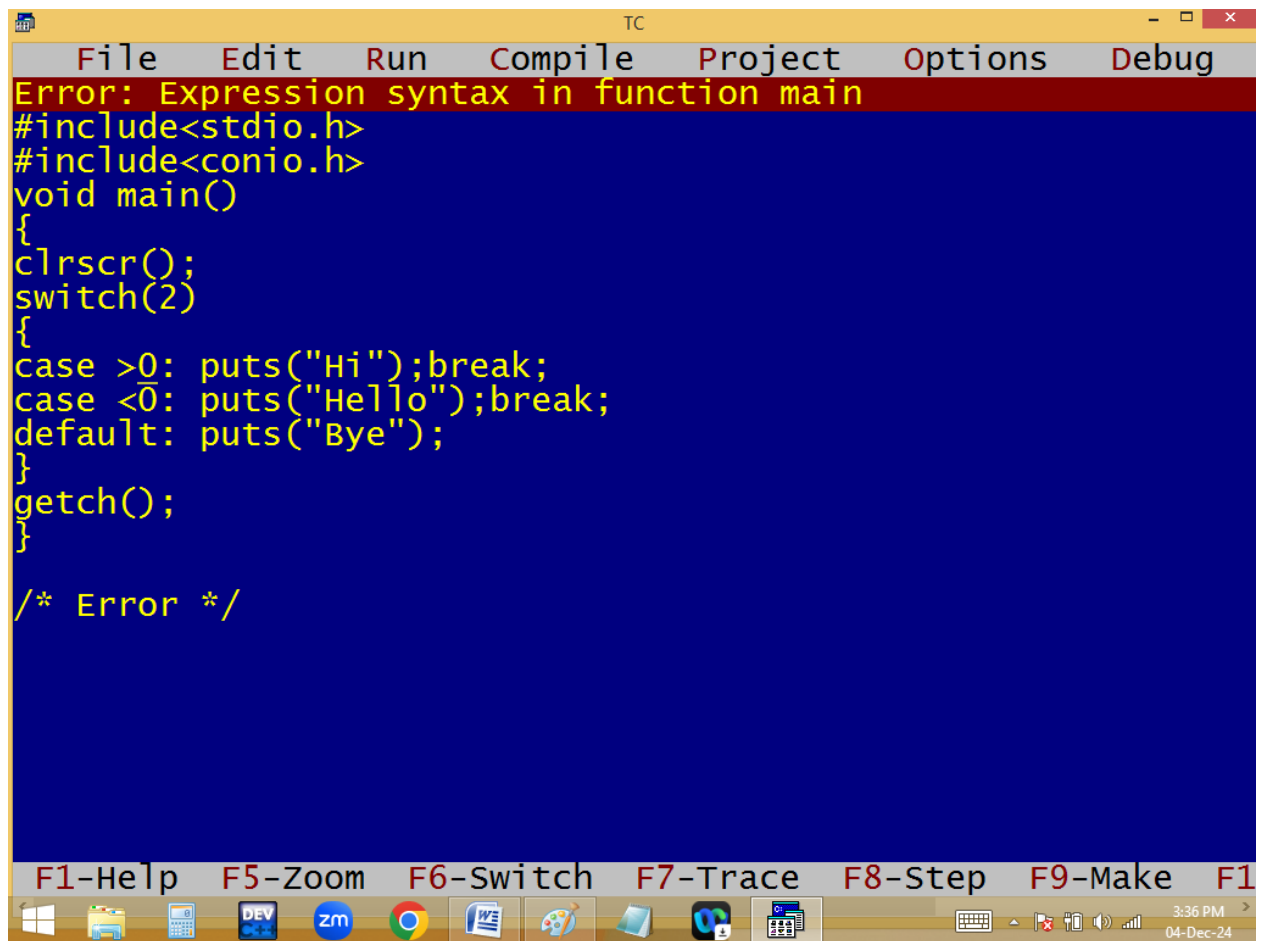


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 14", "Col 9", and "Insert Indent Tab Fill Unindent * E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(0,1)
{
case 0: puts("Hi");break;
case 1: puts("Hello");
}
getch();
}

/* Hello_*/
```

The bottom of the window features a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows various application icons, including the Start button, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and others. The system clock in the bottom right corner displays "3:34 PM" and "04-Dec-24".

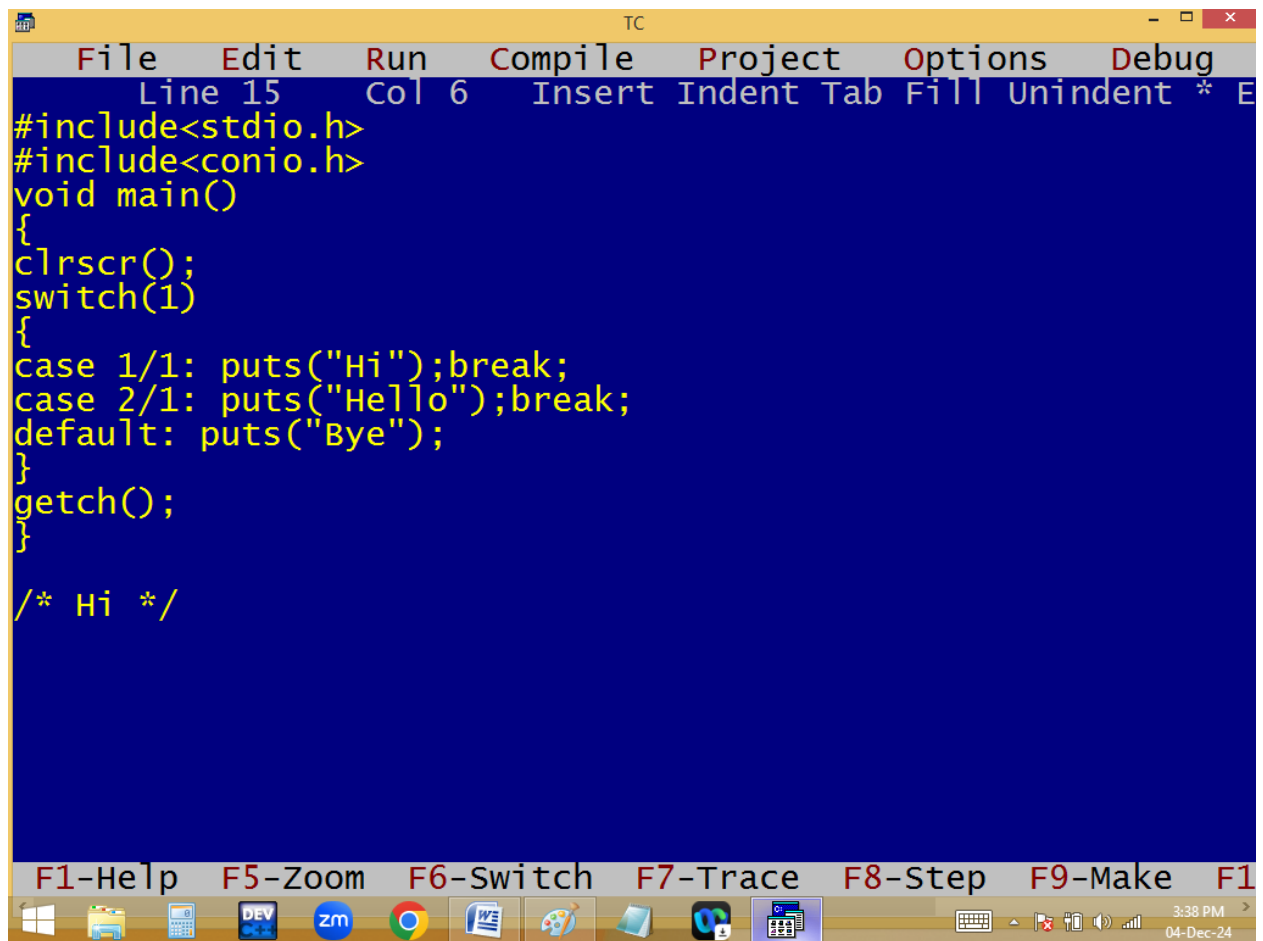


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code area states "Error: Expression syntax in function main". The code being edited is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(2)
{
case >0: puts("Hi");break;
case <0: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Error */
```

Below the code area, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the bottom contains icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and a folder. The system tray on the right shows the time as 3:36 PM and the date as 04-Dec-24.

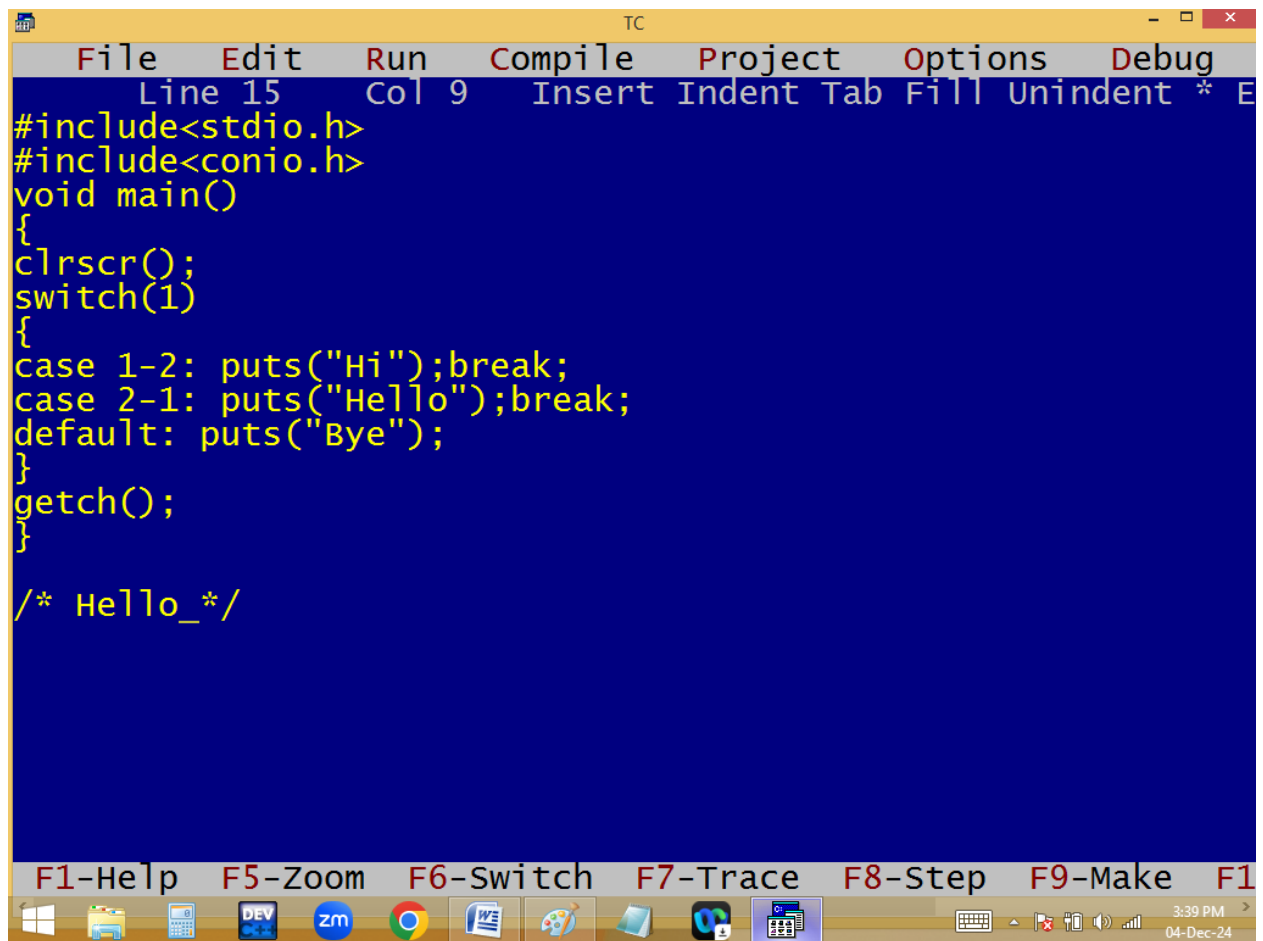


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 15", "Col 6", and "Insert Indent Tab Fill Unindent * E". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
case 1/1: puts("Hi");break;
case 2/1: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Hi */
```

The bottom of the window features a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows the Start button, several application icons (including File Explorer, Calculator, DEV C++, Zoom, Chrome, Word, and Paint), and the system tray with the time "3:38 PM" and date "04-Dec-24".

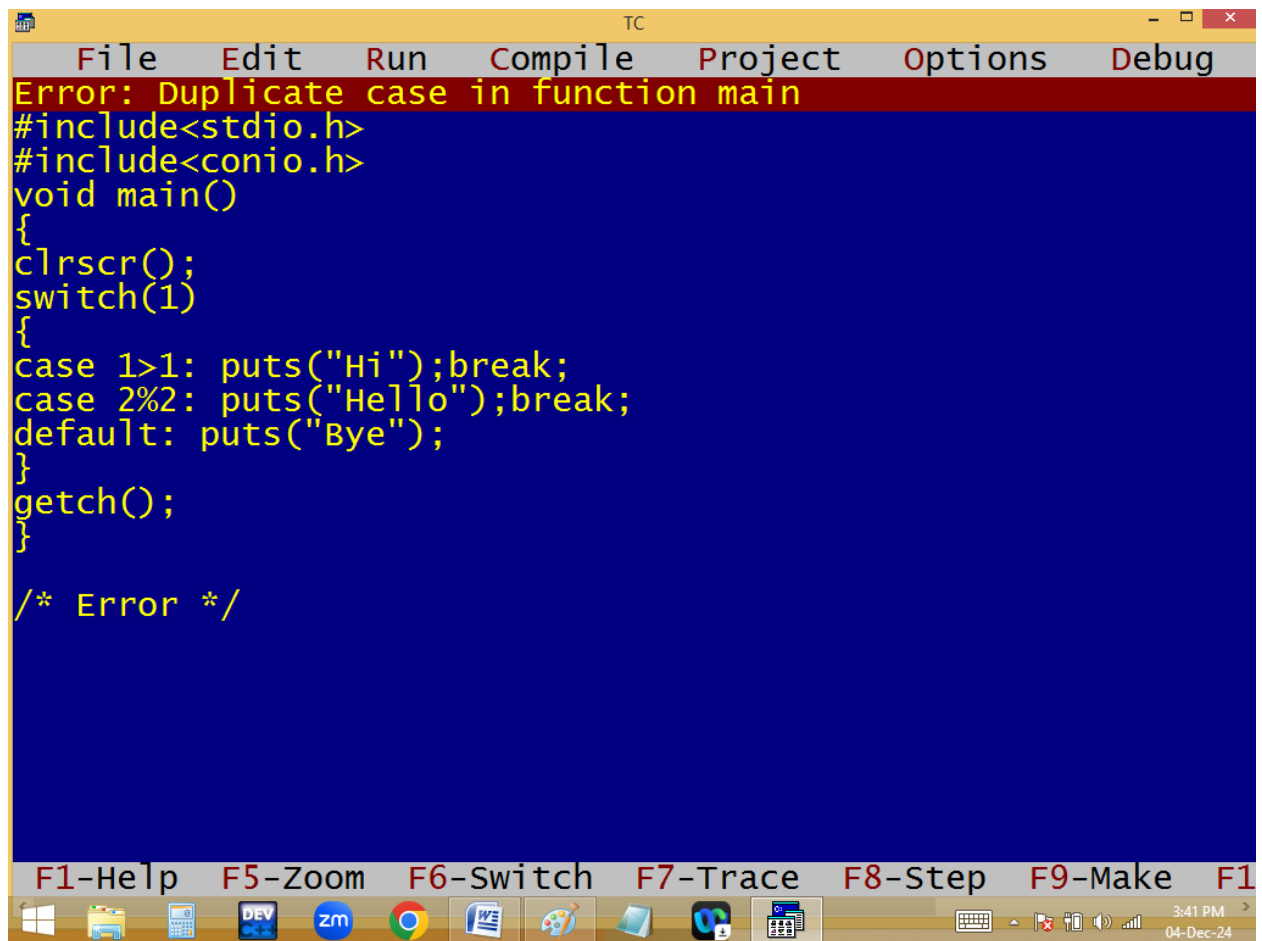


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 15", "Col 9", and "Insert Indent Tab Fill Unindent * E". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
case 1-2: puts("Hi");break;
case 2-1: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Hello_*/
```

The bottom of the window features a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar at the very bottom shows the Start button, several application icons (including File Explorer, Calculator, DEV C++, Zoom, Chrome, Word, and Paint), and the system tray with the time "3:39 PM" and date "04-Dec-24".

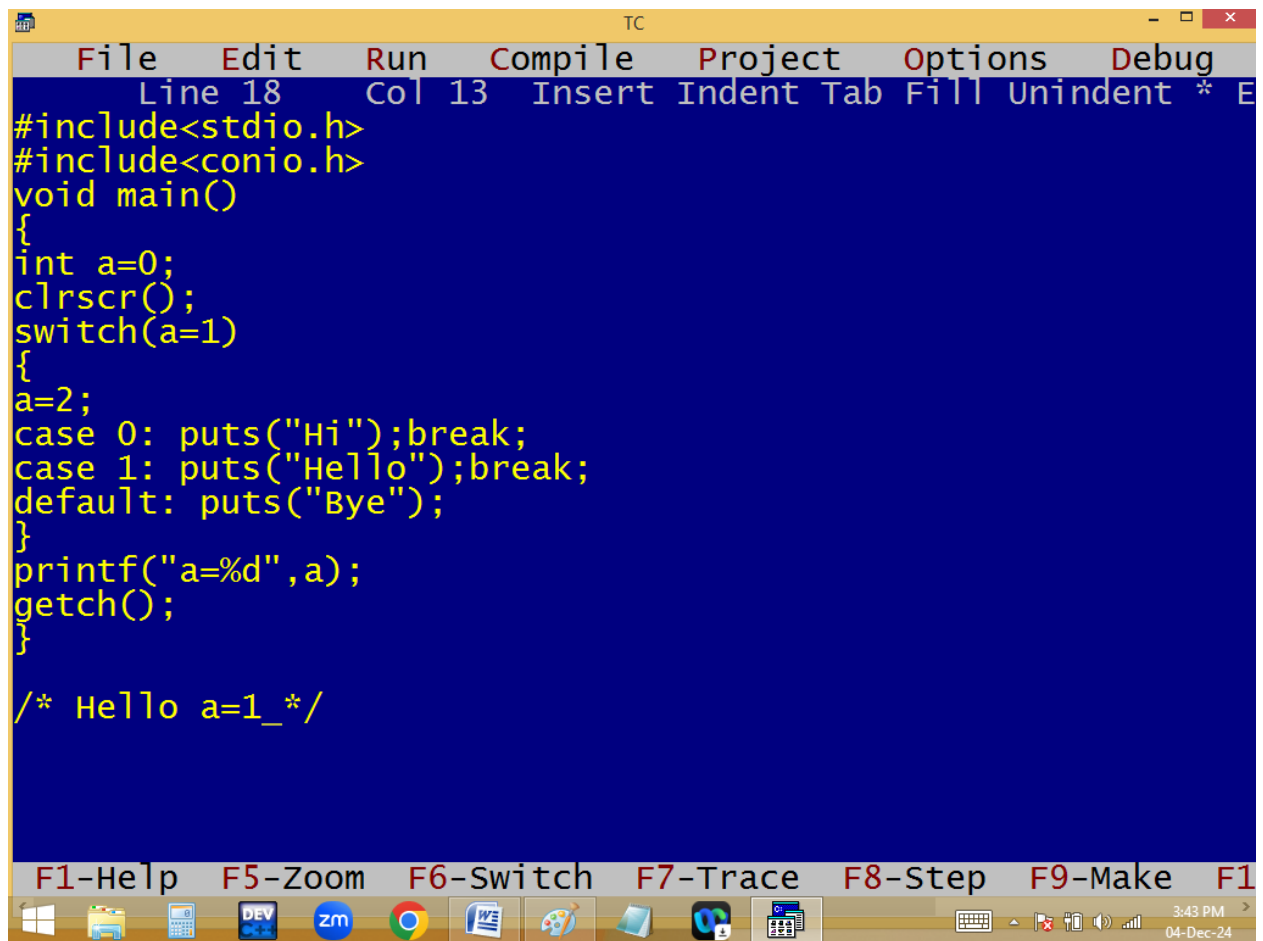


The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states "Error: Duplicate case in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
switch(1)
{
case 1>1: puts("Hi");break;
case 2%2: puts("Hello");break;
default: puts("Bye");
}
getch();
}

/* Error */
```

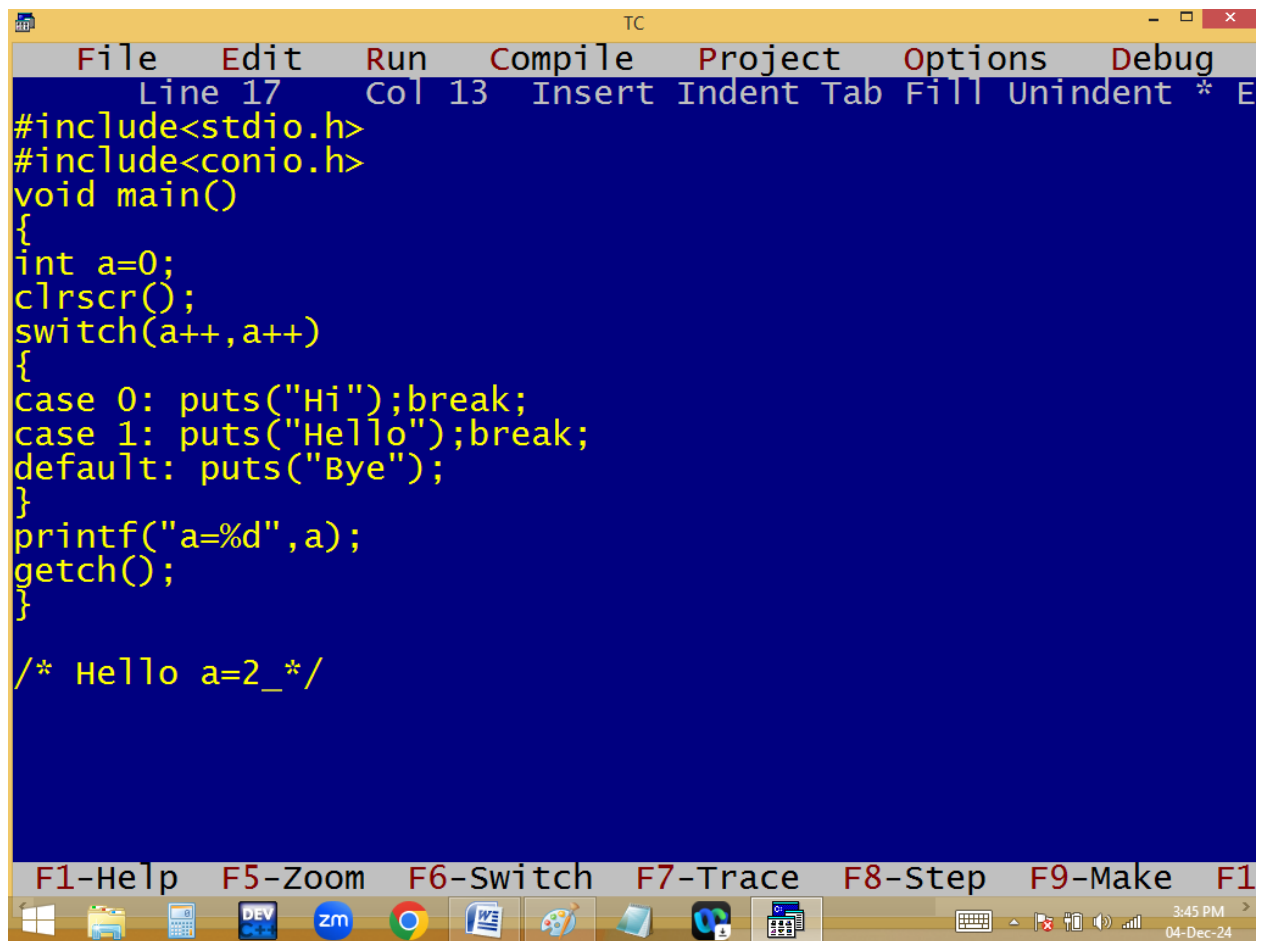
Below the code editor, there is a toolbar with function key shortcuts: "F1-He lp", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F1". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Word, Paint, and VS Code. The system tray on the right shows the time as 3:41 PM and the date as 04-Dec-24.



```
TC
File Edit Run Compile Project Options Debug
Line 18 Col 13 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=1)
{
a=2;
case 0: puts("Hi");break;
case 1: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hello a=1_*/

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run
3:43 PM 04-Dec-24
```

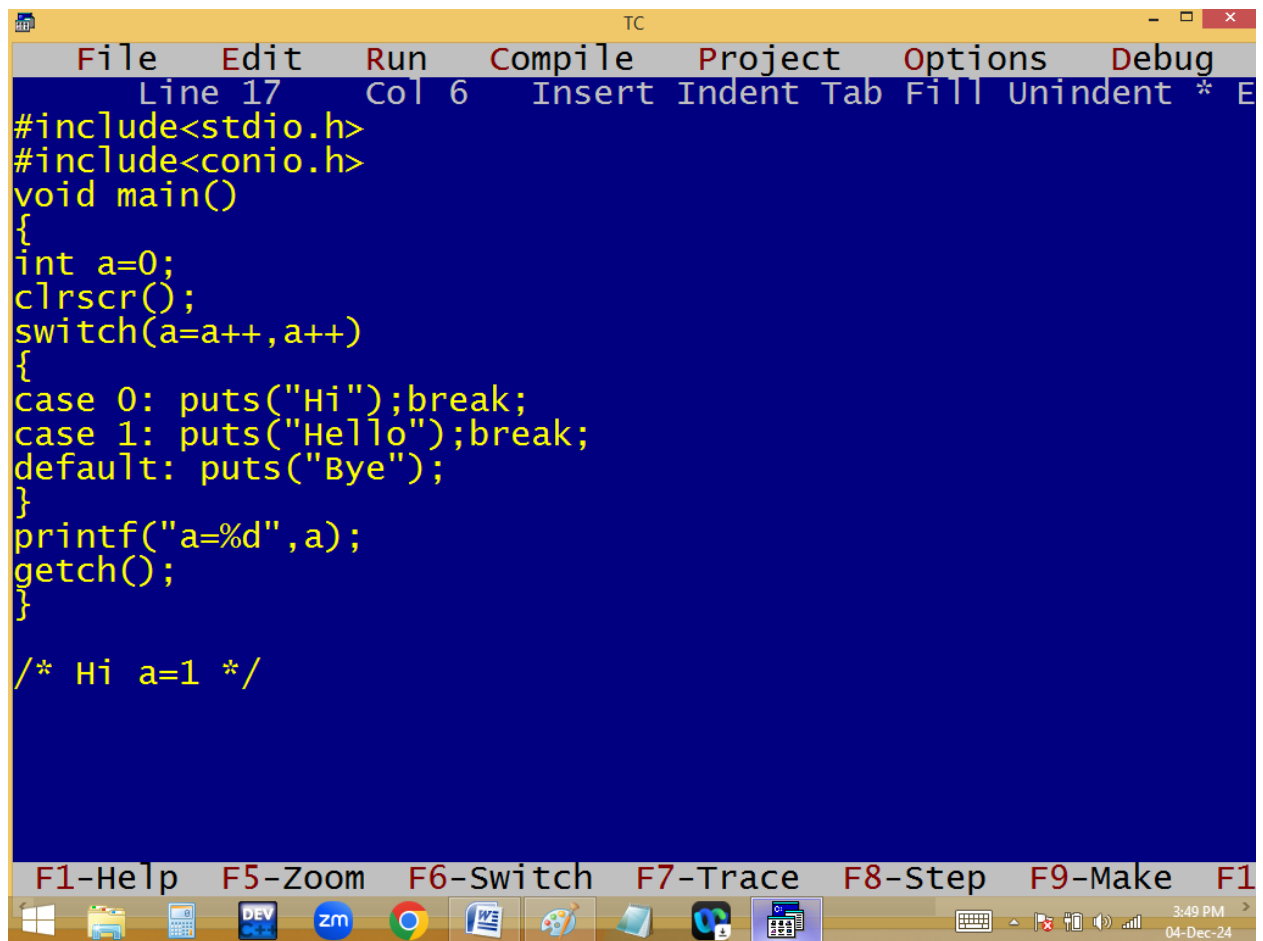


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 17", "Col 13", and lists editing options: "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a++,a++)
{
case 0: puts("Hi");break;
case 1: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hello a=2_*/
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Zoom, Google Chrome, Notepad, Paint, and the Turbo C++ application. The system clock in the bottom right corner shows "3:45 PM" and "04-Dec-24".

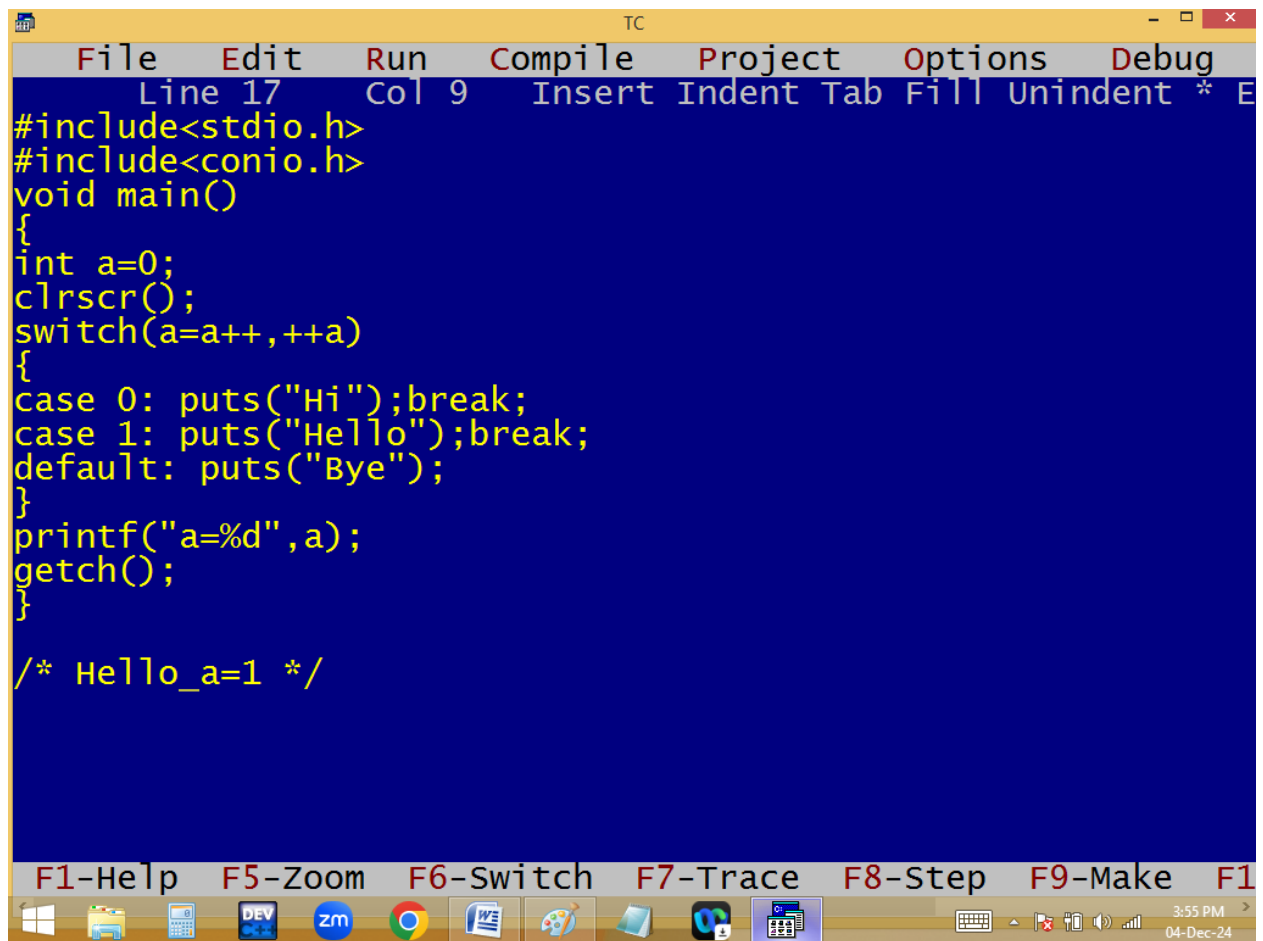


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 17", "Col 6", and "Insert Indent Tab Fill Unindent * E". The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=a++,a++)
{
case 0: puts("Hi");break;
case 1: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hi a=1 */
```

The bottom of the window features a toolbar with icons for various functions and a status bar at the very bottom showing the time "3:49 PM" and the date "04-Dec-24".



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 17", "Col 9", and "Insert Indent Tab Fill Unindent * E". The code is written in yellow text on a blue background. It includes headers for `stdio.h` and `conio.h`, and defines a `main` function. Inside `main`, a variable `a` is initialized to 0, the screen is cleared with `clrscr()`, and a `switch` statement is used to print "Hi", "Hello", or "Bye" based on the value of `a`. The value of `a` is incremented in the `switch` statement. After the `switch`, the value of `a` is printed using `printf`, and a character is read with `getch()`. A comment `/* Hello_a=1 */` is at the bottom of the code. The bottom of the window features a toolbar with function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar at the very bottom shows various application icons and the system clock indicating 3:55 PM on 04-Dec-24.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=0;
clrscr();
switch(a=a++,++a)
{
case 0: puts("Hi");break;
case 1: puts("Hello");break;
default: puts("Bye");
}
printf("a=%d",a);
getch();
}

/* Hello_a=1 */
```